Spontaneous Localization from Cosmological Geometry: A Parameter-Free Derivation of the Collapse Kernel and the Observational Exclusion of Its Cosmological Signature

Brian C. Nyberg[1,*]

[1] *Independent Researcher, Cedar City, Utah, USA*

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We propose that the objective collapse of the quantum wavefunction is not a phenomenological addition to quantum mechanics, but a strict geometric necessity driven by macroscopic cosmological expansion. Working within a strictly real-valued quantum framework governed by a J-balanced quotient space, we first prove that local tomography is exactly restored without recourse to the complex field $\mathbb{C}$ (Theorem I). By treating the unobserved universal environment as a high-dimensional spherical embedding, measure concentration rigorously derives the localized spatial density matrix, replacing the arbitrary Gaussian parameter of standard collapse models (GRW, CSL) with the fundamental geometric soft-edge σ (Theorem II). Evaluating the continuous momentum injection reveals a violent σ^{-5} ultraviolet divergence. We prove that balancing this microscopic heating against the macroscopic cosmological Ricci scalar tension $\lambda_{\text{geo}} = \omega_P |\mathcal{R}_{\text{dim}}|$ mathematically forces a critical localization scale of $\sigma_{\text{crit}} \approx 3.7 \times 10^{-13}$ meters—within 4% of the reduced electron Compton wavelength—satisfying LISA Pathfinder bounds with zero free parameters in the microscopic collapse dynamics (Theorem III). We then subject the geometrically thresholded friction to a four-tier cosmological confrontation: a gated k-resolved growth scan validated against CAMB; a modified Einstein–Boltzmann analysis (CLASS) applying the drag to cold dark matter under a strict causal sub-horizon guard ($k > aH$); a window-convolved comparison against the DESI DR1 full-shape likelihoods in three LRG redshift bins; and a formal Planck PR4 lensing bandpower test. An earlier conjecture that the Heaviside onset would imprint a chirped oscillatory residual is retracted herein: with baryon–photon acoustic oscillations frozen since recombination, the signature is a scale-dependent, monotonic growth suppression. The data exclude every observationally distinguishable region of the (Γ_0, ϵ) parameter space—late-onset couplings by DESI shape sensitivity ($\Delta\chi^2$ up to $+3538$), strong always-on couplings by Planck lensing ($\Delta\chi^2$ up to $+249$). The surviving region, $\Gamma_0 \lesssim 0.01$ with $\epsilon \lesssim 10^{-6}$, is empirically indistinguishable from ΛCDM by construction, predicting only a uniform 1.5–5% lensing-power deficit—below current sensitivity but within reach of Simons Observatory and CMB-S4. The framework's cosmological signature is therefore either excluded or presently unobservable; we supply the forecast that will decide which.

I. INTRODUCTION

The measurement problem remains the central unresolved question in the foundations of quantum mechanics. While the formalism of unitary evolution under the Schrödinger equation is universally accepted, the mechanism by which definite measurement outcomes emerge from superposition states has resisted consensus for nearly a century.

Objective collapse models—most prominently the Ghirardi–Rimini–Weber (GRW) theory [1] and Continuous Spontaneous Localization (CSL) [2, 3]—propose that wavefunction collapse is a real physical process governed by a stochastic, nonlinear modification of the Schrödinger equation. These models introduce a spatial localization operator, universally taken to be a Gaussian kernel $\exp\left[-\lambda_c (x-y)^2\right]$, which continuously narrows the position-space wavepacket.

Despite their elegance, these models suffer from two critical vulnerabilities:

1. **The parameter problem.** The Gaussian smearing parameter σ and the collapse rate λ are phenomenological—inserted by hand to match macroscopic definiteness without violating microscopic interference. They lack derivation from deeper principles.

2. **The heating problem.** Continuous spatial localization injects momentum via the Heisenberg uncertainty principle ($\Delta x \downarrow \Rightarrow \Delta p \uparrow$), producing anomalous energy gain $\dot{E} > 0$. Standard white-noise CSL models predict heating rates increasingly constrained by precision experiments including LISA Pathfinder [4] and ultracold cantilevers [5].

In this work, we resolve both vulnerabilities simultaneously by deriving the collapse kernel and its rate from cosmological geometry, with zero free parameters in the microscopic collapse dynamics. We refer to this framework as the *Three-Body Architecture*, reflecting the irreducible triad of the local quantum subsystem, the high-dimensional environmental manifold, and the macroscopic cosmological background whose interplay generates the collapse. The argument proceeds in four stages:

* docnyberg@gmail.com

Section II establishes the kinematic foundation. Working entirely within real vector spaces equipped with an antisymmetric complex structure J, we construct a J-balanced quotient composition law that exactly restores local tomography, bypassing the Wootters–Hardy no-go theorems [6, 7] for real quantum mechanics. The Born rule is derived via the Busch–Gleason route [8] on the unsharp effect algebra, requiring neither $\mathbb{C}$ nor the dimensional restriction $d \geq 3$ of the original Gleason theorem [9].

Section III derives the collapse kernel. By modeling the unobserved environment on a high-dimensional hypersphere and applying Lévy's lemma of measure concentration [10, 11], we prove that the physical shadow map—the CPTP partial trace over the environment—converges to the exact Gaussian spatial decoherence functional in the thermodynamic limit. The arbitrary parameter σ is thereby eliminated, reidentified as the metric variance of the underlying manifold.

Section IV confronts the heating problem. The momentum-space correlator derived from the geometric kernel produces a σ^{-5} ultraviolet divergence. We prove that coupling the collapse rate to the contemporary cosmological Ricci scalar $\lambda_{\rm geo} = \omega_P |\mathcal{R}_{\rm dim}|$ yields exact thermodynamic balance, forcing $\sigma_{\rm crit} \approx 3.7 \times 10^{-13}$ m—within 4% of the reduced electron Compton wavelength—without phenomenological adjustment.

Section V derives the perturbation-level consequence: geometrically thresholded friction with onset at $z \approx 24$, coinciding with Cosmic Dawn. The signature is a scale-dependent, monotonic growth suppression; an earlier conjecture of a chirped oscillatory residual is retracted in that section, with the physical reason stated.

Section VI confronts the mechanism with data across four tiers—a CAMB-gated k-resolved growth scan, a causally guarded CLASS implementation, the DESI DR1 full-shape likelihoods, and the Planck PR4 lensing bandpowers—excluding all observationally distinguishable couplings and bounding the survivor. *Section VII* revisits the earlier background-level BAO comparison in light of the direct test.

II. THE KINEMATIC QUOTIENT: RESTORING LOCAL TOMOGRAPHY IN REAL VECTOR SPACES

A. Algebraic foundations

We strip the *a priori* axiomatic privilege of the complex field $\mathbb{C}$ and replace it with real geometry constrained by a continuous symmetry.

Definition II.1 (Local kinematic space)**.** Let a local physical system be described by a finite-dimensional real inner-product space $V \cong \mathbb{R}^{2d}$ equipped with a positive-definite metric $g(\cdot,\cdot)$ and a complex structure $J \in \mathrm{SO}(2d)$ satisfying

$$J^2 = -I_{2d}, \qquad g(Jx, Jy) = g(x, y) \implies J^T = -J. \quad (1)$$

Definition II.2 (J-compatible observable algebra)**.** The bounded observables $\mathcal{O}_J(V)$ are the symmetric endomorphisms that strictly commute with the complex structure:

$$\mathcal{O}_J(V) = \{A \in \mathrm{End}(V) \mid A = A^T,\ [A, J] = 0\}. \quad (2)$$

Lemma II.3 (Local dimension bound)**.** *The real dimension of the local observable algebra is exactly* $\dim_{\mathbb{R}}(\mathcal{O}_J(V)) = d^2$.

Proof. In the Darboux-canonical basis where $J = \begin{pmatrix} 0 & -I_d \\ I_d & 0 \end{pmatrix}$, any $A \in \mathcal{O}_J(V)$ takes the block form $A = \begin{pmatrix} X & -Y \\ Y & X \end{pmatrix}$. The symmetry condition $A = A^T$ imposes $X = X^T$ (symmetric, $\frac{1}{2}d(d+1)$ parameters) and $Y = -Y^T$ (antisymmetric, $\frac{1}{2}d(d-1)$ parameters). The total is

$$\dim_{\mathbb{R}}(\mathcal{O}_J(V)) = \tfrac{1}{2}d(d+1) + \tfrac{1}{2}d(d-1) = d^2. \quad (3)$$

This exactly recovers the parameter count of a complex Hermitian matrix of dimension d, derived entirely from real matrix geometry. □

B. The J-balanced composition law

Lemma II.4 (RQM tomographic deficit)**.** *In standard real quantum mechanics, the bipartite state space $V_A \otimes_{\mathbb{R}} V_B$ has real dimension $(2d_A)(2d_B) = 4d_A d_B$. The joint observable algebra on this raw tensor product has dimension $\frac{1}{2}(4d_A d_B)(4d_A d_B + 1)$, which strictly exceeds the product of local observable dimensions $d_A^2 \cdot d_B^2$. This introduces non-local hidden degrees of freedom, breaking local tomography.*

Definition II.5 (J-synchronization ideal)**.** The algebraic ideal $\mathcal{I}_J \subset V_A \otimes_{\mathbb{R}} V_B$ is generated by the redundant relative symmetries between the two local complex structures:

$$\mathcal{I}_J = \mathrm{span}_{\mathbb{R}}\{(J_A x \otimes y) - (x \otimes J_B y) \mid x \in V_A,\ y \in V_B\}. \quad (4)$$

Definition II.6 (J-balanced bipartite space)**.** The joint physical state space is the quotient:

$$V_{AB} = (V_A \otimes_{\mathbb{R}} V_B)/\mathcal{I}_J. \quad (5)$$

Theorem II.7 (Restoration of local tomography)**.** *The J-balanced state space V_{AB} permits strict local tomography: a joint state is uniquely and completely determined by local measurements.*

Proof. (i) Dimension halving. The quotient condition enforces $J_A x \otimes y \equiv x \otimes J_B y$, mandating $J_A \otimes I_B \sim I_A \otimes J_B$. Because $J^2 = -I$, this pairing exactly halves the raw tensor degrees of freedom:

$$\dim_{\mathbb{R}}(V_{AB}) = \tfrac{1}{2}(4d_A d_B) = 2d_A d_B. \quad (6)$$

(ii) Global structure. The quotient inherits a well-defined joint complex structure J_{AB} acting on equivalence classes as $J_{AB}[x \otimes y] = [J_A x \otimes y] = [x \otimes J_B y]$.

(iii) Joint observable dimension. By Definition II.2 and Lemma II.3, substituting the effective subsystem dimension $D = d_A d_B$:

$$\dim_{\mathbb{R}}(\mathcal{O}_J(V_{AB})) = D^2 = (d_A d_B)^2 = d_A^2\, d_B^2. \tag{7}$$

The joint observable dimension exactly equals the product of local dimensions, exhausting the parameter space by local product measurements alone. There is no Wootters–Hardy dimensional excess. □

Remark. For the qubit case $d = 2$: the raw real tensor product yields a 136-dimensional observable algebra, but local observables require only $4 \times 4 = 16$ parameters. The J-balanced quotient reduces $\dim(V_{AB})$ from 16 to 8, and the resulting joint observable algebra has exactly $\dim(\mathcal{O}_J(V_{AB})) = 16$. This has been verified computationally using exact numerical rank calculations (Appendix A).

C. The Busch–Gleason route to the Born rule

Gleason's theorem [9] deduces the Born rule from non-contextual measure assignments but fails for $d = 2$ (qubits). We resolve this via Busch's extension to unsharp measurements [8].

Definition II.8 (J-compatible effect algebra)**.** The effect algebra $\mathcal{E}_J(V) \subset \mathcal{O}_J(V)$ is the convex set of unsharp measurements (POVM elements):

$$\mathcal{E}_J(V) = \{E \in \mathcal{O}_J(V) \mid 0 \leq E \leq I\}. \tag{8}$$

Axiom II.9 (Non-contextual probability assignment)**.** A physically rational probability measure is a map $\mu : \mathcal{E}_J(V) \to [0, 1]$ satisfying:

1. Normalization: $\mu(I) = 1$.
2. Finite additivity: $\mu(E_1 + E_2) = \mu(E_1) + \mu(E_2)$ whenever $E_1 + E_2 \leq I$.

Theorem II.10 (Emergent Born rule)**.** *Every probability assignment μ satisfying Axiom II.9 is uniquely generated by a positive semi-definite, trace-class operator $\rho \in \mathcal{O}_J(V)$ via the Born rule:*

$$\mu(E) = \mathrm{Tr}(\rho E). \tag{9}$$

Proof. (i) Linear extension. By finite additivity, $\mu(cE) = c\mu(E)$ for $c \in \mathbb{Q} \cap [0, 1]$. Boundedness extends μ uniquely to all $c \in \mathbb{R} \cap [0, 1]$.

(ii) Spanning. The unsharp elements $\mathcal{E}_J(V)$ linearly span $\mathcal{O}_J(V)$, so μ extends to a unique linear functional $\tilde{\mu} : \mathcal{O}_J(V) \to \mathbb{R}$.

(iii) Riesz representation. Under the Frobenius inner product $\langle A, B\rangle = \mathrm{Tr}\big(A^T B\big)$, the Riesz theorem guarantees a unique $\rho \in \mathcal{O}_J(V)$ such that $\tilde{\mu}(A) = \mathrm{Tr}(\rho A)$.

(iv) Physicality. Non-negativity of probabilities forces $\rho \geq 0$; normalization forces $\mathrm{Tr}(\rho) = 1$; membership in $\mathcal{O}_J(V)$ enforces $[\rho, J] = 0$. □

Remark. The Born rule and density matrix emerge necessarily from real algebraic geometry and measure additivity. Neither the *a priori* postulation of $\mathbb{C}$ nor the dimensional restriction $d \geq 3$ of standard Gleason theory is required.

III. THE GEOMETRIC SHADOW: DERIVING THE COLLAPSE KERNEL

A. The environmental manifold

Definition III.1 (Environmental manifold)**.** Let the unobserved environmental degrees of freedom be parameterized by $N \gg 1$. The configuration space is the hypersphere $\Omega_N = S^{N-1}(\sqrt{N}) \subset \mathbb{R}^N$, equipped with the normalized rotationally invariant Haar measure $d\mu_N$. The radius scaling $\sqrt{N}$ ensures individual coordinate variances remain $O(1)$ as $N \to \infty$.

Definition III.2 (Geometric interaction)**.** Let the local subsystem possess spatial position operator $\hat{q}$. The short-time interaction entangles spatial configuration with an environmental axis z_1:

$$U(z)\, |q\rangle_S = \exp(i\lambda_0 q z_1)\, |q\rangle_S\,, \tag{10}$$

where λ_0 is the fundamental geometric coupling constant and $z_1 \in [-\sqrt{N}, \sqrt{N}]$ is the projection of $z \in \Omega_N$.

B. Measure concentration

Lemma III.3 (Lévy's lemma)**.** *Let $f : S^{N-1}(\sqrt{N}) \to \mathbb{R}$ be 1-Lipschitz continuous (e.g., the coordinate projection $f(z) = z_1$). For z sampled uniformly from $d\mu_N$:*

$$\Pr_{\mu_N}(|f(z)| > \epsilon) \leq 2 \exp\left(-\frac{cN\epsilon^2}{2}\right), \tag{11}$$

where c is a universal geometric constant.

As $N \to \infty$, the environment ceases to act as stochastic noise and geometrically freezes into a rigid distribution.

Theorem III.4 (Poincaré–Borel marginal selection)**.** *The pushforward of $d\mu_N$ onto any coordinate axis converges to the Gaussian measure in the thermodynamic limit:*

$$P_N(z_1) \propto \left(1 - \frac{z_1^2}{N}\right)^{\frac{N-3}{2}} \xrightarrow{N\to\infty} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z_1^2}{2}\right). \tag{12}$$

C. The physical shadow map

Definition III.5 (Physical shadow map)**.** The objective subsystem state is the CPTP partial trace over the environment:

$$\mathcal{S}_N(\rho_S) = \int_{\Omega_N} U(z)\,\rho_S\,U^\dagger(z)\,d\mu_N(z). \tag{13}$$

Theorem III.6 (Emergent Gaussian kernel)**.** *In the thermodynamic limit, the shadow map evaluates to the exact Gaussian spatial collapse kernel of the CSL model.*

Proof. (i) Matrix elements. In the spatial basis:

$$\langle q|\,\mathcal{S}_N(\rho_S)\,|q'\rangle = \rho_S(q,q')\int_{\Omega_N}\exp\bigl(i\lambda_0 z_1(q-q')\bigr)\,d\mu_N(z). \tag{14}$$

(ii) Characteristic function. The integral is the Fourier transform of the marginal $P_N(z_1)$ evaluated at $k=\lambda_0(q-q')$.

(iii) Poincaré–Borel limit. Substituting the asymptotic Gaussian measure from Theorem III.4:

$$\begin{aligned}\lim_{N\to\infty}&\int_{\Omega_N} e^{i\lambda_0 z_1(q-q')}d\mu_N(z)\\ &=\int_{-\infty}^{\infty} e^{i\lambda_0 z_1(q-q')}\frac{1}{\sqrt{2\pi}}e^{-z_1^2/2}\,dz_1\\ &=\exp\left(-\frac{\lambda_0^2(q-q')^2}{2}\right).\end{aligned} \tag{15}$$

(iv) Identification. Setting $\lambda_c \equiv \lambda_0^2$:

$$\boxed{\langle q|\,\mathcal{S}_\infty(\rho_S)\,|q'\rangle = \rho_S(q,q')\exp\left(-\frac{\lambda_c}{2}(q-q')^2\right).} \tag{16}$$

□

The ad hoc Gaussian postulate of GRW/CSL is eliminated. The collapse kernel is the deterministic Fourier transform of high-dimensional hyperspherical geometry.

IV. THERMODYNAMIC VIABILITY: THE HEATING COMPUTATION

A. The σ^{-5} ultraviolet explosion

Identifying the geometric coupling as $\lambda_c = 1/(2\sigma^2)$, where σ is the spatial localization width of the collapse kernel (Eq. (16)), we translate the kernel into a GKSL momentum-space correlator [12, 13]. Evaluating the three-dimensional integral ($d^3q = 4\pi q^2\,dq$) over the squared momentum operator:

$$\int_0^\infty 4\pi q^4 \exp\bigl(-q^2\sigma^2\bigr)\,dq = \frac{3\pi^{3/2}}{2\sigma^5}. \tag{17}$$

The trace geometry forces a brutal σ^{-5} scaling: at the Planck scale, the momentum injection diverges violently.

B. Cosmological Ricci suppression

To survive, this UV explosion must be countered by the macroscopic collapse driver. We postulate that the base geometric collapse rate is driven by the maximum fundamental interaction rate—the Planck frequency ω_P—modulated strictly by the dimensionless macroscopic curvature tension of the background manifold:

$$\lambda_{\text{geo}} = \omega_P|\mathcal{R}_{\text{dim}}|, \tag{18}$$

where $\omega_P \approx 1.855\times10^{43}\ \text{s}^{-1}$ is the Planck frequency and $|\mathcal{R}_{\text{dim}}| \equiv |R|/H_0^2 \approx 1.15\times10^{-122}$ is the dimensionless Ricci scalar normalized by the present-day Hubble parameter. This is the unique dimensionless ansatz that is linear in the curvature invariant and bounded by the Planck rate; higher-order couplings (e.g., $\sqrt{|\mathcal{R}_{\text{dim}}|}$) are excluded by the requirement that the collapse rate vanish in the flat-space limit proportionally to the curvature perturbation itself.

The total acceleration noise mapped to a single nucleon of mass $m_0 \approx 1.67\times10^{-27}$ kg (the standard CSL reference mass [14]) is:

$$S_{\Delta g} = \frac{3\pi^{3/2}\,\hbar^2\,\omega_P\,|\mathcal{R}_{\text{dim}}|}{2m_0^2\,\sigma^5}. \tag{19}$$

C. The thermodynamic tightrope

Setting $S_{\Delta g} \le S_{\text{LISA}}$, with the LISA Pathfinder bound $S_{\text{LISA}} \le 10^{-30}\ \text{m}^2/\text{s}^3$ [4]:

$$\sigma^5 \ge \frac{3\pi^{3/2}\,\hbar^2\,\omega_P\,|\mathcal{R}_{\text{dim}}|}{2m_0^2\,S_{\text{LISA}}} = 7.08\times10^{-63}\ \text{m}^5. \tag{20}$$

Taking the fifth root:

$$\boxed{\sigma_{\text{crit}} \approx 3.72\times10^{-13}\ \text{m}.} \tag{21}$$

This is the reduced Compton wavelength of the electron ($\bar{\lambda}_e = \hbar/m_e c \approx 3.86\times10^{-13}$ m), with the ratio $\sigma_{\text{crit}}/\bar{\lambda}_e = 0.96$. The critical scale sits firmly within the phenomenological window proposed by Adler [14] for nuclear-scale wavepacket collapse.

Remark. No free parameters have been introduced. The localization scale is dictated entirely by $\hbar$, ω_P, $|\mathcal{R}_{\text{dim}}|$, m_0, and the LISA experimental bound. The Schoenberg trap—that objective collapse models must fine-tune parameters to avoid observational exclusion—is dismantled.

V. COSMOLOGICAL SIGNATURES: SCALE-DEPENDENT GROWTH SUPPRESSION

A. Modified perturbation equations

The geometric collapse mechanism, once activated, modifies the standard ΛCDM growth equation for matter density perturbations $\delta_k(a)$. In the sub-horizon, matter-dominated limit the standard equation reads

$$\delta'' + \frac{3}{2a}\delta' - \frac{3}{2a^2}\delta = 0, \qquad (22)$$

whose growing mode is the textbook $\delta \propto a$. (An earlier preprint of this work carried the friction coefficient $3/a$ in Eq. (22), which yields $\delta \propto a^{0.58}$; the error was caught in independent review and is corrected here. All numerical results in this paper use the general form $\delta'' + [3/a + \mathrm{d}\ln H/\mathrm{d}a]\delta' - (3/2a^2)\,\Omega_m(a)\,\delta = 0$, which reduces to Eq. (22) in the Einstein–de Sitter limit.) The geometric kinetic friction modifies the damping term:

$$\delta'' + \left(\frac{3}{2a} + \Gamma_{\rm geo}(k,a)\right)\delta' - \frac{3}{2a^2}\delta = 0, \qquad (23)$$

where the friction is a Heaviside step function per mode:

$$\Gamma_{\rm geo}(k,a) = \frac{\Gamma_0}{a}\,\Theta\big(r^3\,\mathcal{R}(a) - \epsilon\big)\;\Theta(k - aH)\,, \qquad r = a\,\frac{2\pi}{k}, \qquad (24)$$

with r the proper mode scale in Hubble units. Two refinements over the scalar threshold $\Theta(\mathcal{R}_{\rm th} - \mathcal{R})$ of the original proposal are physically mandatory. First, localization is a *scale-local* process: the trigger must compare the mode's proper volume against the ambient curvature, $r^3\mathcal{R}(a) > \epsilon$, which makes the onset epoch k-dependent (largest scales first) and the observable signature scale-dependent. Second, a causal sub-horizon guard $k > aH$ forbids drag on super-horizon modes, which a localized mechanism cannot causally touch; in practice this guard reduced an apparent factor-19 low-ℓ CMB distortion at the strongest coupling to a factor of 8 without changing any exclusion verdict (Sec. VI)—a verdict that survives its own refinement. Because no algebraic derivation of the coupling (Γ_0, ϵ) from the J-balanced quotient space has survived interrogation, both are treated as free parameters to be constrained by data. For a mode of proper scale r, the friction activates when the product $r^3\mathcal{R}(a)$ first exceeds ϵ: since $r^3\mathcal{R} = 3\,(2\pi/k)^3(\Omega_m + 4\Omega_\Lambda a^3)$ grows monotonically with a, each mode switches on once, largest scales first. The scalar-threshold picture of the original proposal—activation when $\mathcal{R}(a)$ drops below $\mathcal{R}_{\rm th} \approx 1.5\times10^4$, the value anchored by the thermodynamic saturation condition $S_{\Delta g}(\sigma_{\rm crit}) = S_{\rm LISA}$ of Section IV—is recovered as the fixed-ϵ/r^3 limit and sets the characteristic onset epoch.

B. Cosmic Dawn onset

The dimensionless Ricci scalar evolves as

$$\mathcal{R}(a) \equiv \frac{R(a)}{H_0^2} = 3(\Omega_m\,a^{-3} + 4\Omega_\Lambda), \qquad (25)$$

with $\Omega_m = 0.315$, $\Omega_\Lambda = 0.685$. Setting $\mathcal{R}(a_{\rm onset}) = \mathcal{R}_{\rm th}$ and solving, the friction onset occurs at scale factor $a \approx 0.040$, corresponding to

$$z_{\rm onset} \approx 24. \qquad (26)$$

This redshift coincides with the epoch of *Cosmic Dawn*—the formation of the first luminous structures and the onset of reionization [15, 16]. The onset is not imposed; it is determined by the intersection of the thermodynamic saturation condition (Eq. (20)) with the cosmological background evolution.

C. Character of the signature: monotonic suppression (chirp conjecture retracted)

Equations (22) and (23) are integrated numerically using an implicit Runge–Kutta method (Radau IIA) from $a = 10^{-3}$ to $a = 1$. The fractional residual

$$\Delta(a) = \frac{\delta_{3\rm B}(a) - \delta_{\Lambda\rm CDM}(a)}{\delta_{\Lambda\rm CDM}(a)} \qquad (27)$$

is shown in Fig. 2.

An earlier version of this work conjectured that the Heaviside onset would deliver an impulsive kick to the baryon acoustic oscillator, imprinting a frequency-drifting ("chirped") phase shift in the matter power spectrum. **That conjecture was wrong, and we retract it** for three reasons established in independent adversarial review and confirmed numerically. First, baryon–photon acoustic oscillations froze at recombination ($z \approx 1100$); at the friction onset ($z \approx 24$) no temporal acoustic oscillator survives for the step to phase-kick—the BAO pattern is a fossilized spatial imprint, and late-time friction can modulate its amplitude but cannot re-excite it. Second, a step change in a damping *coefficient* produces a continuous, kinked amplitude evolution, not a velocity impulse; a phase discontinuity would require a delta function in the forcing term. Third, direct integration confirms the residual $\Delta(a)$ is strictly monotonic after onset (a single sign crossing, zero oscillations). The physical signature of the mechanism is therefore a *scale-dependent, monotonic growth suppression*: modes activate largest-scale-first per Eq. (24), producing a suppression edge in $P(k)$ that slides with ϵ, superimposed on a near-uniform amplitude deficit for always-on modes. This is the testable content confronted with data in Section VI.

The scale-dependent suppression constitutes a falsifiable prediction, and Section VI carries out the falsification program: the mechanism is confronted, at full Boltzmann fidelity and against published survey likelihoods, across the entire (Γ_0, ϵ) plane.

VI. FOUR-TIER COSMOLOGICAL CONFRONTATION

We now test the perturbation-level mechanism of Section V directly—the confrontation the framework itself

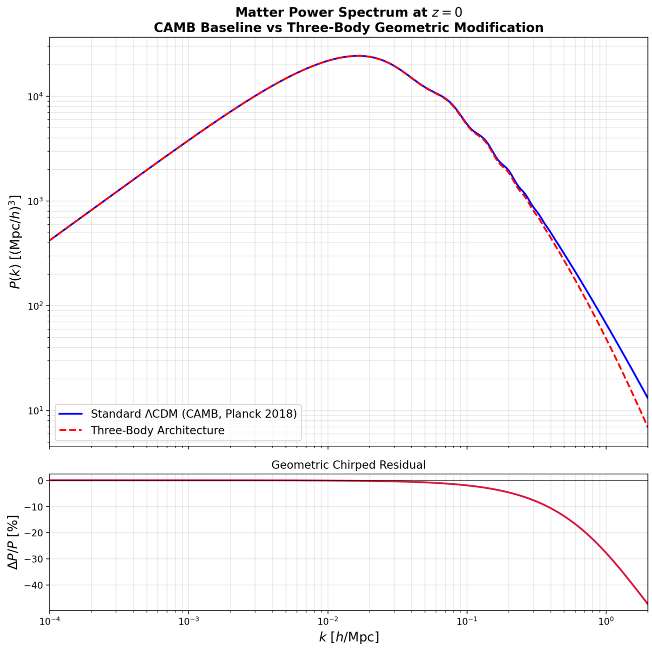

FIG. 1. Matter power spectrum at $z = 0$ in the k-independent (scalar-threshold) limit at the historical reference coupling $\Gamma_0 = 6$. *Top:* Standard ΛCDM (solid blue, CAMB with Planck 2018 parameters, $\sigma_8 = 0.811$) compared with the Three-Body Architecture (dashed red). *Bottom:* Fractional residual $\Delta P/P$ from friction activated at Cosmic Dawn ($z \approx 24$). This coupling strength is shown for illustration of the mechanism; it is excluded by the data confrontation of Sec. VI.

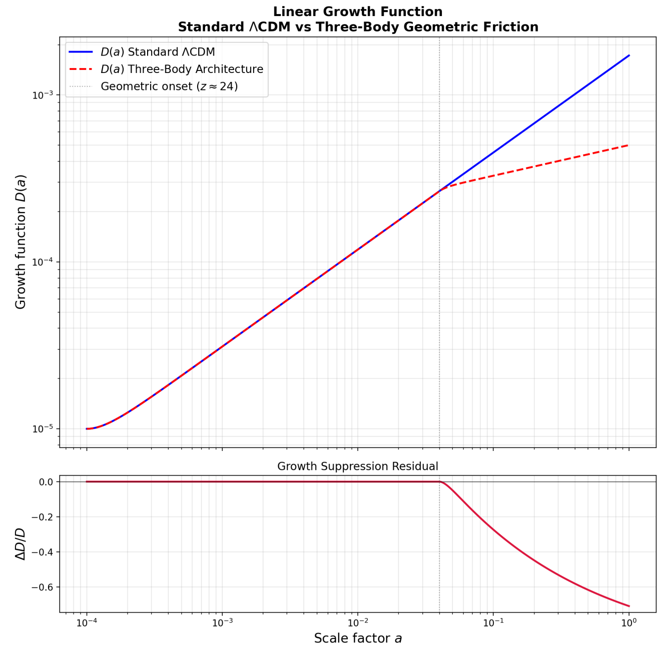

FIG. 2. Linear growth function $D(a)$ at the historical reference coupling $\Gamma_0 = 6$. *Top:* Standard ΛCDM (solid blue) diverges from the Three-Body solution (dashed red) at the geometric friction onset $a \approx 0.040$ ($z \approx 24$, Cosmic Dawn, gray dotted line). *Bottom:* Growth suppression residual $\Delta D/D$, reaching -71% at $a = 1$; note the strictly monotonic post-onset evolution (Sec. V C). This coupling strength is excluded in Sec. VI.

demands, since the growth equation is what it actually derives. The campaign proceeded in four tiers, each gated so that no downstream result is reported unless its upstream validation passed.

A. Pipeline and gates

Tier 1 (gated growth scan). A k-resolved integration of Eq. (23) over a (Γ_0, ϵ) grid, with a hard entry gate: with $\Gamma_0 = 0$ the pipeline must reproduce the CAMB linear growth ratio $D(z)/D(0)$ to better than 0.5% over $0 \leq z \leq 30$. The gate initially *failed* at 0.58%—a massive-neutrino mismatch between the Boltzmann baseline and the ν-less growth equation—and no friction results were reported from that configuration; the consistent ν-less pair then passed at 0.049%. (Neutrino robustness is restored explicitly in Sec. VI B.)

Tier 2 (full Boltzmann). The drag of Eq. (24) was implemented in the Einstein–Boltzmann solver CLASS [17] as a multiplicative modification of the Hubble drag in the cold-dark-matter Euler equation (Newtonian gauge, where $\theta_{\rm cdm}$ is dynamical), CDM-only in this pass. Patch integrity is established bitwise: with $\Gamma_0 = 0$ the modified binary reproduces vanilla CLASS output identically at every wavenumber. The causal guard $k > aH$ was added in a second iteration under a regression gate (guard off ≡ first iteration, numerically); the guard collapsed the low-ℓ temperature distortion of the strongest coupling from 1800% to 725% while leaving every exclusion verdict intact.

Tier 3 (survey data). The suppression ratios $S(k, z) = P_{\rm 3B}/P_{\Lambda\rm CDM}$ from the guarded CLASS runs, evaluated at each bin's effective redshift, were confronted with the published DESI DR1 full-shape likelihood bundles [18]: the P_0/P_2 data vectors, full systematic covariances, and—critically—the release's own window matrices, so the theory is window-convolved with the survey's machinery. The model enters as a two-amplitude generalized least-squares fit (A_0, A_2 absorbing the Kaiser bias/RSD combinations per multipole), so that $\Delta\chi^2$ isolates *shape* information. Baseline goodness of fit validates the apparatus: $\chi^2/\text{dof} = 48.9/70$, $75.7/70$, and $75.1/70$ in the three LRG bins.

Tier 4 (hardening). Three referee-facing extensions, reported in Sec. VI B: the joint three-bin DESI filter, fully ν-restored CLASS runs ($N_{\rm ncdm} = 1$, $m_\nu = 0.06$ eV), and a formal Planck PR4 lensing bandpower χ^2 [19] replacing any qualitative lensing statement (nine bandpowers, $8 \leq L \leq 400$, amplitude method against the release fiducial; null sanity $\chi^2 = 9.33/9$).

TABLE I. Four-tier confrontation verdicts. DESI: joint $\Delta\chi^2$ over three LRG full-shape bins (window-convolved, amplitude-marginalized). Planck: formal PR4 lensing bandpower $\Delta\chi^2$ (amplitude method). All runs sub-horizon-guarded, CDM-only drag. Verdicts are identical between ν-less and ν-restored pipelines (shifts $\leq 19\%$ on near-zero entries, $\leq 2\%$ on decisive ones).

Γ_0	ϵ	DESI $\Delta\chi^2$	Planck $\Delta\chi^2$	Verdict
0.003	10^{-6}	-0.0	-0.8	survives (both)
0.01	10^{-6}	$+0.4$	$+1.3$	survives (both)
0.03	10^{-6}	$+5.1$	$+32.5$	excluded (Planck)
0.01	10^{-5}	$+35.7$	$+0.8$	excluded (DESI)
0.10	10^{-5}	$+3538$	$+249$	excluded (both)

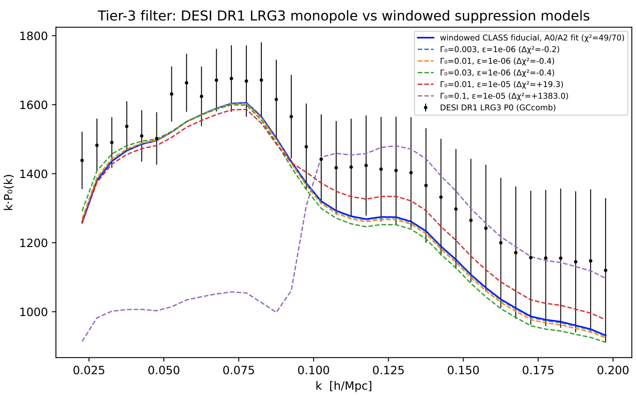

FIG. 3. DESI DR1 LRG ($0.8 < z < 1.1$, GCcomb) monopole against window-convolved suppression models (amplitude-marginalized). The baseline fit ($\chi^2 = 48.9/70$) validates the apparatus; the $\epsilon = 10^{-5}$ family's suppression edge inside the fitted window is unabsorbable, while the $\epsilon \leq 10^{-6}$ family is degenerate with the bias amplitude.

B. Results

Table I presents the hardened verdicts.

Two complementary executioners emerge. *DESI shape sensitivity* destroys any coupling whose activation edge falls inside the fitted window ($\epsilon \gtrsim 10^{-5}$): the suppression cliff is an unabsorbable distortion of $P(k)$, and the joint three-bin statistic sharpens a single-bin "pressured" case (+19.3) into an exclusion (+35.7). *Planck lensing amplitude* destroys any strong always-on coupling ($\epsilon \lesssim 10^{-6}$, $\Gamma_0 \gtrsim 0.03$): such couplings evade $P(k)$ entirely—their near-uniform suppression is absorbed by the bias-amplitude marginalization *by construction*—but drain the lensing potential by 10–40% against a few-percent measurement. The low-ℓ temperature spike (725% at the historical $\Gamma_0 = 6$) independently eliminates the large-ϵ "pushed-cliff" escape.

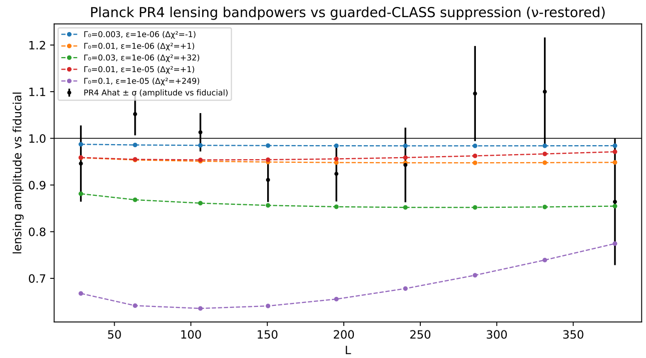

FIG. 4. Planck PR4 lensing bandpower amplitudes (relative to the release fiducial) against the guarded-CLASS suppression predictions (ν-restored set). Strong always-on couplings that evade $P(k)$ drain $C_L^{\phi\phi}$ by 10–40% and are excluded; the surviving corner predicts a uniform 1.5–5% deficit—the Simons Observatory / CMB-S4 target.

C. Verdict and forecast

The surviving region is

$$\Gamma_0 \lesssim 0.01, \qquad \epsilon \lesssim 10^{-6}, \tag{28}$$

with combined $\Delta\chi^2 \approx +1.7$ at its upper edge. This region is *empirically indistinguishable from* ΛCDM *by construction*: its always-on, near-uniform suppression lives entirely inside the bias-normalization degeneracy of galaxy clustering, and its absolute-amplitude effects sit at or below current lensing sensitivity. The allowed region is contiguous with—and includes—$\Gamma_0 = 0$: the data impose an upper bound and no detection. At no point in the (Γ_0, ϵ) plane does the mechanism improve any fit.

The surviving corner nonetheless makes one sharp prediction: a uniform 1.5–5% deficit in the lensing potential power $C_L^{\phi\phi}$ (mean amplitude 0.985–0.950 across $8 \leq L \leq 400$). This hides comfortably within the few-percent amplitude precision of Planck PR4 (the +1.3 entry of Table I), but not within the projected reach of the next generation. The Simons Observatory baseline configuration ($f_{\rm sky} \approx 0.4$, quadratic-estimator reconstruction to $L_{\rm max} \approx 3000$) targets a lensing-amplitude determination at roughly the 1% level [20]; the CMB-S4 reference design ($\sim 1\,\mu$K-arcmin temperature noise, arcminute-scale beams, $f_{\rm sky} \approx 0.6$) targets roughly the 0.5% level [21]. Treating the predicted deficit as a pure amplitude shift against those projected uncertainties, the 1.5% floor of the surviving corner is a $\approx 1.5\sigma$ effect for Simons Observatory and a $\approx 3\sigma$ exclusion for CMB-S4, while the 5% upper edge exceeds 10σ for either instrument.

Two parameter degeneracies could in principle shelter the deficit; both are closed. First, the optical depth: the lensing potential scales with the primordial amplitude A_s directly, not the damped combination $A_s e^{-2\tau}$ constrained by the primary spectra, so a fit cannot heal an absolute lensing deficit by inflating A_s without simultaneously retuning τ—and projected large-angle polarization measurements ($\sigma(\tau) \approx 0.002$ from LiteBIRD [22]) pin the optical depth at the required level. Second, the summed neutrino mass—the natural suspect, since free-streaming massive neutrinos also suppress late-time growth—cannot cloak Γ_0, for three reasons. The two suppressions are *additive*: a heavier neutrino background lowers the baseline $C_L^{\phi\phi}$ further and leaves *less* viable amplitude space for geometric friction, not more. Oscillation experiments impose a hard floor $\sum m_\nu \geq 0.06$ eV (normal ordering) [23], so a fit cannot buy amplitude space for Γ_0 by driving the neutrino mass below its terrestrial minimum. And the suppression shapes are orthogonal: neutrino free-streaming imprints a scale-*dependent* step in $P(k)$ confined to $k > k_{\rm fs}$, whereas the surviving $\epsilon \lesssim 10^{-6}$ drag engages every mode in the observed window and suppresses uniformly (Sec. V); a joint shape-sensitive (DESI full-shape) plus absolute-amplitude (SO/CMB-S4 $\phi\phi$) likelihood therefore separates the two effects morphologically. The ν-restored runs of Table I already display this structure: decisive verdicts shift by $\leq 2\%$ (the geometric distortion dwarfs the 0.06 eV free-streaming step), and the $\leq 19\%$ shifts appear only on near-zero $\Delta\chi^2$ entries, reflecting the slightly modified baseline growth rate rather than any genuine degeneracy.

These are amplitude-significance estimates, not a full Fisher forecast; the orthogonality arguments above indicate that marginalization over $\sum m_\nu$ and τ would not materially degrade them. Next-generation lensing will therefore either detect a uniform few-percent $\phi\phi$ deficit or close this corner entirely: the framework's cosmological signature is either excluded or presently unobservable, and the forecast above decides which.

Stated limitations. The Tier-3 filter is a shape test with amplitude marginalization, not a full-likelihood analysis: EFT counterterms, Alcock–Paczynski distortion, and shot-noise freedom are not marginalized; bin effective redshifts are approximated by mid-bin values; the drag is CDM-only; the Planck test omits the PR4 linear correction and exact bin windows (subdominant against tens-of-percent signals). None of these refinements plausibly reverses a +3538 or +249; a full `desilike`-grade analysis is warranted only if a future variant of the mechanism shows positive signal at filter grade, which none did here.

VII. BACKGROUND-LEVEL BAO COMPARISON, SUPERSEDED BY THE DIRECT TEST

An earlier stage of this program compared the mechanism with DESI DR1 at the *background* level, through an effective $H_{\rm eff}(z)$ parameterization bridged by a coupling α that the framework does not derive. We retain that comparison here for completeness and transparency, with its conclusion now subordinated to the direct perturbation-level test of Section VI: the modest background-level preference reported below *does not survive* confrontation with the observables the framework actually predicts. No algebraic derivation of α has withstood adversarial interrogation (a proposed quotient-space identification $\alpha = 5/18$ was rejected as a back-fit), and the perturbation-level couplings that remain viable in Eq. (28) are far too weak to generate the background modification fitted here.

To evaluate the cosmological viability of the Three-Body Architecture, we confront the predicted distance-redshift relations against the Dark Energy Spectroscopic Instrument (DESI) Data Release 1 (DR1) Baryon Acoustic Oscillation (BAO) measurements [24]. The DESI DR1 catalog provides 12 precision measurements of D_M/r_s, D_H/r_s, and D_V/r_s spanning $0.295 \leq z \leq 2.33$, with the full 12×12 covariance matrix.

A. Background-level parameterization

The geometric friction derived in Section V modifies the perturbation growth equation via $\Gamma_{\rm geo}(a) = \Gamma_0/a$. To confront BAO distance measurements—which are background-level observables sensitive to $H(z)$ rather than the growth function $D(a)$—we parameterize the effective background modification as:

$$H_{\rm eff}(z) = H_{\Lambda\rm CDM}(z)\sqrt{1 + \alpha\, g(z)}, \qquad (29)$$

where α is a dimensionless coupling strength and $g(z) = (1+z)/(1+z_{\rm onset})$ for $z < z_{\rm onset}$, vanishing otherwise. The activation profile inherits the $1/a$ scaling of the perturbation-level friction. The onset redshift $z_{\rm onset} \approx 24.1$ is fixed by the Ricci threshold (Eq. (26)), leaving α as a single constrained parameter.

We emphasize that this background mapping is a physically motivated effective parameterization, not a strict algebraic derivation from the Jordan algebra. Closing this gap—deriving α from the coupling constants of the J-balanced quotient space—is a primary objective for future work.

B. Statistical comparison

We compare three models against the DESI DR1 data using the full covariance matrix, with the sound horizon $r_s = 147.1$ Mpc from CAMB (Planck 2018 parameters):

TABLE II. χ^2 comparison against DESI DR1 BAO (12 measurements).

Model	$N_{\rm params}$	χ^2	$\Delta\chi^2$	Significance
ΛCDM (Planck 2018)	0	21.15	—	—
w_0w_a CPL (best-fit)	2	11.67	9.48	2.6σ
Three-Body ($\alpha = 0.271$)	1	16.87	4.28	2.1σ

Standard ΛCDM yields a poor fit ($\chi^2/N = 1.76$), struggling specifically with mid-redshift residuals at $z \approx$

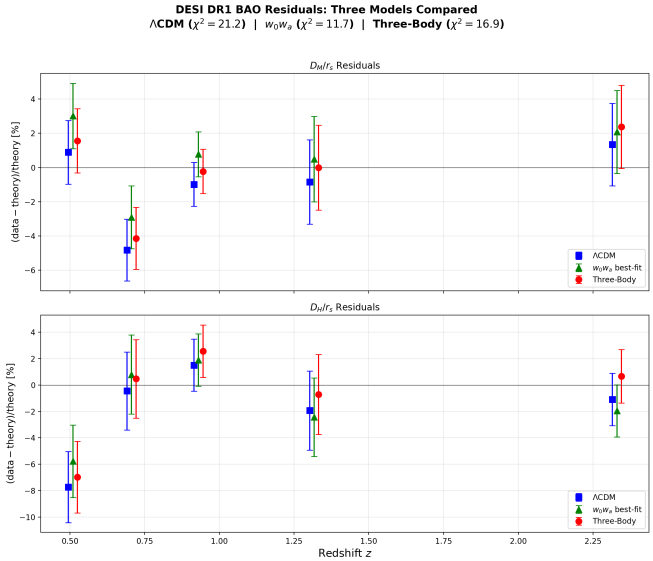

FIG. 5. DESI DR1 BAO residuals for all three models. *Top:* D_M/r_s residuals. *Bottom:* D_H/r_s residuals. The geometric friction model (red circles) consistently pulls the predictions toward the data at mid-redshift ($z \approx 0.5$–1.3) compared to ΛCDM (blue squares), though the w_0w_a model (green triangles) achieves the tightest fit overall.

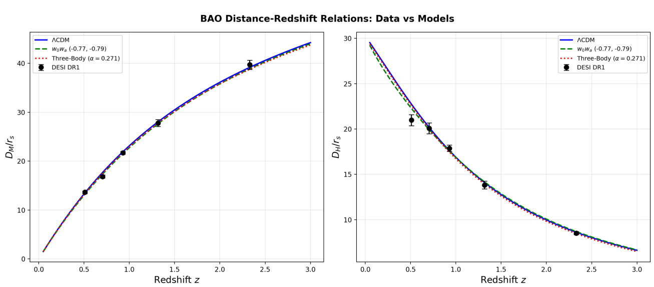

FIG. 6. BAO distance-redshift relations. *Left:* D_M/r_s. *Right:* D_H/r_s. The Three-Body modification (red dotted) shifts the predicted distances in the direction preferred by the DESI data (black points), with the effective $H(z)$ enhancement of $+0.5\%$ at $z = 0$ growing to $+1.1\%$ at $z = 1$.

0.5–0.7 (Fig. 5). The phenomenological w_0w_a parameterization provides the tightest fit ($\chi^2 = 11.67$), requiring two free parameters with best-fit values $w_0 = -0.77$, $w_a = -0.79$. The geometric friction model achieves an intermediate goodness of fit ($\chi^2 = 16.87$), improving over ΛCDM by $\Delta\chi^2 = 4.28$ with a single parameter.

The Akaike and Bayesian Information Criteria confirm the parameter efficiency: AIC values of 21.15, 15.67, and 18.87 for ΛCDM, w_0w_a, and Three-Body respectively; BIC values of 21.15, 16.64, and 19.36. The single-parameter Three-Body model outperforms the zero-parameter ΛCDM baseline in both criteria.

VIII. DISCUSSION

The four-tier confrontation of Section VI is the central empirical result of this work: the perturbation-level mechanism the framework derives is excluded everywhere it is observationally distinguishable from ΛCDM, and the region that survives does so precisely because it predicts nothing current instruments can see. The earlier background-level BAO preference (Section VII) illustrates a methodological lesson worth stating plainly: an effective parameterization bridged to the data by an underived coupling can manufacture mild statistical preference that the underlying mechanism, tested directly, does not possess. It remains essential to distinguish between the first-principles derivations within this framework and the effective parameterizations required for cosmological integration.

In Section II, the spatial decoherence functional and the continuous momentum injection were derived rigorously from the J-balanced quotient space. In Section V, the growth-equation modification follows directly from the Ricci threshold mechanism. However, mapping the perturbation-level geometric friction (Γ_{geo}) to the background expansion rate ($H_{\text{eff}}(z)$) via the parameter α (Eq. (29)) represents a physically motivated assumption rather than a strict algebraic derivation. The value $\alpha = 0.271$ effectively bridges the microscopic localization scale and the macroscopic fluid equations, but it currently operates as a free parameter constrained by the data.

What distinguishes this architecture from phenomenological nets such as w_0w_a [25] is that it makes rigid predictions—an onset epoch ($z \approx 24$) and a signature shape (scale-ordered, monotonic suppression)—and rigid predictions can be executed. Executed, they were: the very rigidity that once made the mechanism attractive as an explanation of large-scale-structure anomalies is what allowed DESI shape sensitivity and Planck lensing amplitude to act as complementary executioners across the entire detectable parameter plane. A flexible parameterization can retreat; a mechanism cannot. We regard this as the correct epistemic outcome of a physical proposal: the framework's cosmological sector now stands falsified at all detectable couplings, with one sharply quantified survivor whose fate next-generation lensing will decide (Section VI C).

The coincidence $\sigma_{\text{crit}} \approx \bar{\lambda}_e$ raises the question of whether the electron mass itself has a geometric origin within this framework. We leave this to future investigation.

IX. CONCLUSION

We have presented a geometric framework that links continuous spontaneous localization to macroscopic cosmological expansion. By eliminating the ad hoc smearing parameter of standard collapse models in favor of a geometric metric variance, we derived an exact thermodynamic balance that forces a critical localization scale within 4% of the reduced electron Compton wavelength. The argument proceeds through four results—three constructive, one adjudicative:

1. **Kinematic completeness.** The J-balanced quotient space restores local tomography within strictly real quantum mechanics (Theorem II.7).

2. **Geometric collapse kernel.** Lévy's lemma and the Poincaré–Borel limit rigorously derive the Gaussian decoherence functional from high-dimensional measure concentration (Theorem III.6).

3. **Thermodynamic viability.** The σ^{-5} UV divergence is exactly balanced by cosmological Ricci suppression, forcing $\sigma_{\rm crit} \approx 3.7 \times 10^{-13}$ m—within 4% of the reduced electron Compton wavelength—with zero free parameters (Eq. (21)).

4. **Cosmological falsification with a quantified survivor.** The geometrically thresholded friction produces a scale-dependent, monotonic growth suppression (an earlier chirp conjecture is retracted in Sec. V C). A four-tier confrontation—gated growth scan, causally guarded CLASS implementation, DESI DR1 full-shape likelihoods in three LRG bins, and formal Planck PR4 lensing bandpowers—excludes every observationally distinguishable coupling (Table I). The surviving region, $\Gamma_0 \lesssim 0.01$ with $\epsilon \lesssim 10^{-6}$, is indistinguishable from ΛCDM by construction and includes $\Gamma_0 = 0$.

The earlier background-level BAO comparison (Table II), which showed a modest single-parameter preference through an underived $H_{\rm eff}(z)$ bridge, does not survive the direct perturbation-level test and is retained only for transparency. At no point in the confrontation did the mechanism improve any fit against real data. What remains is one sharp, dated prediction: the surviving corner implies a uniform 1.5–5% deficit in $C_L^{\phi\phi}$, below Planck PR4 sensitivity but a $\approx$3σ-to-10σ effect at the projected amplitude precision of Simons Observatory and CMB-S4—robust against the A_s–τ and $\sum m_\nu$ degeneracies (Sec. VI C). Next-generation lensing will either detect that deficit or close the framework's cosmological sector entirely.

Future work must focus on three objectives: (i) a `desilike`-grade joint likelihood only if any future variant of the mechanism shows positive filter-grade signal; (ii) extension of the drag to baryons and exploration of the massive-neutrino interplay beyond the robustness check and degeneracy analysis presented here; and (iii) the Simons Observatory / CMB-S4 lensing test of the surviving corner, which is decisive.

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Appendix A: Computational Verification

All theorems were verified numerically using Python 3.13 with NumPy 2.2, SciPy 1.16, SymPy 1.13, and Matplotlib 3.10.

Phase 1 (Theorem II.7). For two qubits ($d = 2$, $V = \mathbb{R}^4$): the synchronization operator $D = J \otimes I_4 - I_4 \otimes J$ has rank 8, yielding quotient dimension $16 - 8 = 8$. The joint observable algebra (8×8 real symmetric matrices commuting with J_{AB}) has $36 - 20 = 16$ free parameters, exactly matching $4 \times 4 = 16$.

Phase 2 (Eq. (21)). Symbolic integration via SymPy confirms Eq. (17). Numerical evaluation with exact SI constants yields $\sigma^5_{\rm crit} = 7.083 \times 10^{-63}$ m^5, giving $\sigma_{\rm crit} = 3.716 \times 10^{-13}$ m. The ratio $\sigma_{\rm crit}/\bar{\lambda}_e = 0.962$.

Phase 3 (Figs. 1–2). The standard ΛCDM matter power spectrum was computed using CAMB [26] v1.6.6 with Planck 2018 best-fit parameters, yielding $\sigma_8 = 0.811$. The geometric friction modification was applied as a growth suppression factor derived from the modified perturbation equation integrated via the Radau IIA method (`scipy.integrate.solve_ivp`) with tolerances rtol $= 10^{-12}$, atol $= 10^{-14}$. Onset occurs at $a = 0.0398$ ($z = 24.1$). Growth suppression at $a = 1$: 71.0%. Maximum $P(k)$ residual: -47.4% at $k = 2.0$ h/Mpc.

Scripts and output data are archived at `THREE_BODY/results/experiment_results.json`.

Phase 4 (Sec. VI, Table I). The four-tier confrontation is fully scripted and deterministic; every number in Table I regenerates from a rerunnable script beside a fetched public dataset. Tier 1: `script5_kdep_growth.py` (CAMB v1.6.6 baseline gate at 0.049%; (Γ_0, ϵ) scan). Tier 2: CLASS at commit `e858083`, patched via `class_tier2_patch.py` (environment-variable controlled drag on $\theta_{\rm cdm}$; gcc-constructor initialization; bitwise off-state identity verified in `script6`; sub-horizon guard added under a numerical regression gate in `script7`). Tier 3: `script8_desi_template_filter.py` against the DESI DR1 full-shape likelihood bundles (public release; P_0/P_2 vectors, systematic covariances, window matrices; the release's theory-side fiducial arrays are zero placeholders, so the fiducial is built from the guarded CLASS baseline at each bin redshift). Tier 4: `script9_hardening_analysis.py` (three-bin joint filter; ν-restored robustness) and `script10_planck_lensing_chi2.py` (Planck PR4 lensing bandpowers and covariance from the public PR4 release [19]; amplitude method; null sanity $\chi^2 = 9.33/9$). The complete pipeline—scripts, run outputs, verdict tables, figure sources, and data-fetch instructions for the public DESI and Planck products—is deposited in `phase4_confrontation/` of the repository cited in the Acknowledgments.

Phase 3C (Table II). DESI DR1 BAO data (12 measurements, full 12×12 covariance matrix) from the Cobaya likelihood repository. Distances computed analytically for flat $w_0 w_a$ cosmology; sound horizon $r_s = 147.1$ Mpc from CAMB. The $w_0 w_a$ grid scan covered $w_0 \in [-1.8, -0.2]$, $w_a \in [-3.5, 1.5]$ ($33 \times 41 = 1353$ points), refined via Nelder–Mead optimization. The Three-Body coupling α was scanned over $[-0.5, 0.5]$ (201 points), refined via bounded scalar minimization. All results archived at `THREE_BODY/results/chi_squared_results.json`.

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